

工作经历

- 腾讯科技（上海）有限公司
- 多模态应用研究员
- 优图实验室神农研究中心
2024年8月 - 现在

项目经历

- **LUCY: Linguistic Understanding and Control Yielding Early Stage of Her**
 - 搭建文本音频并行推理范式的基于多码本的端到端语音模型架构。
 - 增加模型多轮对话和工具调用功能，以batch-parallel解码的方式提升回复智商。
 - 构建情绪感知与情感响应数据，以提升模型基于用户情感生成带感情的回复的能力。
- **VITA-Audio: Fast Interleaved Cross-Modal Token Generation for Efficient Large SLM**
 - 基于单码本编解码器的交错推理范式的端到端语音模型中加入MTP以加速解码时音频token的输出速度。
 - 基于开源数据构建10万小时的音频数据集和5B文本数据，模型ASR/TTS/SQA能力提升到SOTA。
- **VITA-QinYu: Expressive Spoken Language Model for Role-Playing and Singing**
 - 在交错范式的端到端语音模型上引入多码本编解码器，使模型学习音频中的副语言信息，生成有表现力的语音。
 - 构建12K小时自然对话数据，2.6K小时角色扮演数据，1.2K小时哼唱数据，支持模型的对应能力。

教育背景

- **University of Illinois at Urbana-Champaign**
- Champaign, IL
- Ph.D. in Electrical and Computer Engineering; GPA: 3.97
 - M.S in Electrical and Computer Engineering; GPA: 3.96
 - B.S. in Computer Science & Electrical Engineering; GPA: 3.90
- 2022年1月 - 2024年 5月
2019年8月 - 2021年12月
2015年 8月 - 2019年 5月

专业技能

- 软件架构
 - 语言
- Fairseq, ESPNet, SpeechBrain, HuggingFace
中文(Fluent), 英语(TOEFL110), 日语(N1)

论文发表

- **VITA-Audio: Fast Interleaved Cross-Modal Token Generation for Efficient Large SLM**
Zuwei Long, Yunhang Shen, Chaoyou Fu, Heting Gao, Lijiang Li, Peixian Chen, Mengdan Zhang, Hang Shao, Jian Li, Jinlong Peng, Haoyu Cao, Ke Li, Rongrong Ji, Xing Sun
NeurIPS, 2025
- **VITA-1.5: Towards GPT-4o Level Real-Time Vision and Speech Interaction**
Chaoyou Fu, Haojia Lin, Xiong Wang, Yi-Fan Zhang, Yunhang Shen, Xiaoyu Liu, Haoyu Cao, Zuwei Long, Heting Gao, Ke Li, Long Ma, Xiwu Zheng, Rongrong Ji, Xing Sun, Caifeng Shan, Ran He
NeurIPS, 2025
- **Speech-Based Phonetic Transcript Metric**
Heting Gao, Mark Hasegawa-Johnson, Chang D Yoo
IEEE TASLP, 2025
- **SyncDiff: Diffusion-Based Talking Head Synthesis with Bottlenecked Temporal Visual Prior for Improved Synchronization**
Xulin Fan*, Heting Gao*, Ziyi Chen, Peng Chang, Mei Han, Mark Hasegawa-Johnson
WACV, 2025
- **G2PU: Grapheme-to-Phoneme Transduction with Speech Units**
Heting Gao, Mark Hasegawa-Johnson, Chang D Yoo
ICASSP, 2024
- **Speech Self-Supervised Learning Using Diffusion Model Synthetic Data**
Heting Gao, Kaizhi Qian, Junrui Ni, Chuang Gan, Mark A. Hasegawa-Johnson, Shiyu Chang, Yang Zhang
ICML, 2024

- **Mitigating the Exposure Bias in Sentence-Level Grapheme-to-Phoneme (G2P) Transduction**
Eunseop Yoon, Hee Suk Yoon, Dhananjaya Gowda, SooHwan Eom, Daehyeok Kim, John Harvill, Heting Gao, Mark Hasegawa-Johnson, Chanwoo Kim, Chang D Yoo
Interspeech, 2023
- **Listen, Decipher and Sign: Toward Unsupervised Speech-to-Sign Language Recognition**
Liming Wang, Junrui Ni, Heting Gao, Jialu Li, Kai Chieh Chang, Xulin Fan, Junkai Wu, Mark Hasegawa-Johnson, Chang Yoo
ACL (Findings of the Association for Computational Linguistics), 2023
- **WavPrompt: Towards Few-Shot Spoken Language Understanding with Frozen Language Models**
Heting Gao, Junrui Ni, Kaizhi Qian, Yang Zhang, Shiyu Chang, Mark Hasegawa-Johnson
Interspeech, 2022
- **Unsupervised Text-to-Speech Synthesis by Unsupervised Automatic Speech Recognition**
Junrui Ni, Liming Wang, Heting Gao, Kaizhi Qian, Yang Zhang, Shiyu Chang, Mark Hasegawa-Johnson
Interspeech, 2022
- **ContentVec: An Improved Self-Supervised Speech Representation by Disentangling Speakers**
Kaizhi Qian, Yang Zhang, Heting Gao, Junrui Ni, Cheng-I Lai, David Cox, Mark Hasegawa-Johnson, Shiyu Chang
ICML (International Conference on Machine Learning), 2022
- **Domain Generalization for Language-Independent Automatic Speech Recognition**
Heting Gao, Junrui Ni, Yang Zhang, Kaizhi Qian, Shiyu Chang, Mark Hasegawa-Johnson
Frontiers in Artificial Intelligence, 2022
- **Seamless equal accuracy ratio for inclusive CTC speech recognition**
Heting Gao, Xiaoxuan Wang, Sunghun Kang, Rusty Mina, Dias Issa, John Harvill, Leda Sari, Mark Hasegawa-Johnson, Chang D Yoo
Speech Communication, 2022
- **Zero-shot Cross-Lingual Phonetic Recognition with External Language Embedding**
Heting Gao, Junrui Ni, Yang Zhang, Kaizhi Qian, Shiyu Chang, Mark Hasegawa-Johnson
Interspeech, 2021
- **Hierarchical multi-armed bandits for discovering hidden populations**
Suhansanu Kumar, Heting Gao, Changyu Wang, Kevin Chen-Chuan Chang, Hari Sundaram
ASONAM (IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining), 2019

实习经历

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- **PAII Lab** Palo Alto, CA
Research Scientist Internship *May 2023 – Aug 2023*
 - Apply latent diffusion to generate lip movement given a speech segment.
 - Add CTC loss to audio encoder to learn better alignment between speech and phoneme.
 - Incorporate AV-HuBERT encoder and decoder to improve audio-visual synchronization.
 - **MIT-IBM Watson AI Lab** Cambridge, MA
Research Scientist Internship *May 2022 – Aug 2022*
 - Extract WAV2VEC2/HuBERT/XLSR labels using FAIRSEQ to condition the training of speech synthesizing models such as WavENET and PRODIFF.
 - Generate synthetic speech to augment pretraining of WAV2VEC2/HuBERT models for improved speech representation on low-resource languages.
 - **Trova Inc.** Champaign, IL
Software Engineer *May 2021 – Aug 2021*
 - Develop a search engine for users' cloud services (GDrive, Slack, Dropbox, etc.)
 - Write micro-service to communicate search in cloud service providers according search queries.
 - Implement rule-based and BERT-based intent classification for search queries.
 - Design and implement evaluation script to evaluate the overall search engine performance.